

MTD 367 iOS Application Development

- ~~Lesson #1 - Building Apps That Matter and Swift~~
- ~~Lesson #2 - Dealing With Swift Types And SwiftUI~~
- ~~Lesson #3 - Doubling Down On SwiftUI~~
- ~~Lesson #4 - All To Do With SwiftData~~
- ~~Lesson #5 - Streaming Multimedia Swiftly~~
- **Lesson #6 - Putting The Map On Your App (Today)**

? Questions About Lesson #5

- Learn how to show images, play audio or play video in a Swift app

? Questions About Virtual Lab #5

Improve your To-Do List app!

- Let User add their own images and videos in their tasks.

Today's Agenda

- ☐ CoreLocation
- ☐ MapKit
- ☐ ARKit

What Is Core Location?

- Apple's framework for **working with location data**
- Helps apps know:
 - Where the user is
 - How fast they're moving
 - Which direction they're heading



Why Maps & AR?

- Many modern apps combine:
 - Maps to show **where**
 - AR to show **what** at that location
- Great for:
 - Navigation
 - Games
 - Education
 - Tourism apps



What Is MapKit?

- Apple's framework for displaying maps
- Works with:
 - Standard maps
 - Satellite
 - Hybrid views
- Supports:
 - Pins (annotations)
 - User location
 - Routes & overlays

What Is ARKit?

- Apple's framework for Augmented Reality
- Uses device camera + sensors
- Places virtual objects in the real world



Hands-On #7

Improve your To-Do List app!

- Let Users add Location and

✓ Summary

- ✓ CoreLocation
- ✓ MapKit
- ✓ ARKit

Reminder

- TMA 2 is due on 22/8/2025
- ECA is due on 14/10/2025

Virtual Lab #6 (20/9/2025)

Improve your To-Do List app!

? Questions?

I'd love to hear how the class went or how I can improve.

Feel free to email me at:

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 All course materials are available at:

www.github.com/iluzdaf/mtd367

Appendix

CoreLocation

```
import CoreLocation
let locationManager = CLLocationManager()
locationManager.requestWhenInUseAuthorization()
// or
locationManager.requestAlwaysAuthorization()
// Fetch location
if let location = locationManager.location {
    print("User is at \(location.coordinate.latitude), \(location.coordinate.longitude)")
}
// Use delegate methods for updates
class MyLocationDelegate: NSObject, CLLocationManagerDelegate {
    func locationManager(_ manager: CLLocationManager, didUpdateLocations locations: [CLLocation]) {
        if let location = locations.first {
            print("Updated location: \(location.coordinate.latitude), \(location.coordinate.longitude)")
        }
    }
}
```

MapKit

Shows a map centered on San Francisco:

```
import MapKit
struct MapView: View {
    @State private var region = MKCoordinateRegion(
        center: CLLocationCoordinate2D(latitude: 37.7749, longitude: -122.4194),
        span: MKCoordinateSpan(latitudeDelta: 0.05, longitudeDelta: 0.05)
    )

    var body: some View {
        Map(coordinateRegion: $region).frame(height: 300)
    }
}
```


ARKit

Places a small blue box in AR space:

```
import ARKit
struct ARViewContainer: UIViewRepresentable {
    func makeUIView(context: Context) -> ARView {
        let arView = ARView(frame: .zero)
        let box = MeshResource.generateBox(size: 0.1)
        let material = SimpleMaterial(color: .blue, isMetallic: false)
        let modelEntity = ModelEntity(mesh: box, materials: [material])
        let anchor = AnchorEntity(world: SIMD3(x: 0, y: 0, z: -0.5))
        anchor.addChild(modelEntity)
        arView.scene.anchors.append(anchor)
        return arView
    }
}
```